SOFTWARE ENGINEERING MINI-PROJECT

Test Case Update for Test Plan

Real-Time Chat Application Using Socket Programming

Deliverable 4 – Functional and Non-Functional Test Case Update

1. Purpose

This document provides the test-case update to be incorporated into the Test Plan. It ensures sufficient coverage of functional and non-functional requirements and satisfies the project requirement for at least 7–10 test cases.

2. Test Case Design

Each test case identifies the SRS requirement being validated, the test condition, procedure, expected result, and test type. Positive, negative, non-functional, and security scenarios are included.

Test ID	Feature	SRS Trace	Type	Procedure	Expected Result
TC-01	Registration	FR-01	Functional	Enter valid unique registration data and submit.	A new account is created successfully.
TC-02	Valid Authentication	FR-02	Functional	Enter valid registered credentials.	User is authenticated and allowed to access protected chat.
TC-03	Invalid Authentication	FR-03	Functional/Security	Enter incorrect credentials.	Login is rejected and protected access is not granted.
TC-04	Socket Connection	FR-05	Functional	Connect a client to the running server.	Persistent socket/WebSocket connection is established.
TC-05	Send Message	FR-07	Functional	User A sends a valid text message.	Message is accepted for transmission.
TC-06	Receive and Route	FR-08, FR-09, FR-19	Functional	Send a message from User A to User B.	Server routes the message and User B displays the correct message.
TC-07	Timestamp/Order	FR-10, FR-11	Functional	Send several messages sequentially.	Each message has a timestamp and messages appear chronologically.
TC-08	Multiple Users	FR-12, FR-16	Functional	Connect several clients simultaneously.	Clients remain active and online-user list reflects active users.
TC-09	Disconnect Handling	FR-15, FR-20	Functional	Terminate one client connection unexpectedly.	System handles the disconnect and other users remain available.
TC-10	Invalid Message	FR-17, FR-18	Functional/Negative	Attempt to send an empty or invalid message.	Message is rejected and not processed as a normal message.
TC-11	Message Delivery Time	NFR-01, NFR-10	Non-Functional	Measure time from send to receipt under normal expected load.	Measured time is within the 1-second target.
TC-12	Concurrent Connections	NFR-02	Non-Functional	Connect multiple clients concurrently and exchange messages.	Active sessions remain functional without unnecessary termination.
TC-13	Availability/Recovery	NFR-03	Non-Functional	Disconnect/reconnect a client while the server remains operational.	Service remains available and client can reconnect.
TC-14	Reliability	NFR-04	Non-Functional	Send a sequence of valid messages during a stable connection.	Each valid message is delivered once to intended recipient(s).

TC-15	Usability	NFR-05	Non-Functional	Ask a first-time user to connect, send a message, and log out.	User completes the basic workflow without technical training.
TC-16	Input Validation Security	NFR-09, SEC-02	Security	Submit malformed/unexpected input.	Input is rejected or safely handled.
TC-17	Unauthenticated Access	SEC-01, SEC-03	Security	Attempt protected chat operation without authentication.	Access is denied.
TC-18	Unauthorized Administration	SEC-04	Security	Attempt administrative operation without authorization.	Operation is denied.

3. Security Validation Cases

ID	Security Area	SRS Trace	Procedure	Expected Result
SV-01	Authentication Bypass	SEC-01, SEC-03	Access protected chat without authentication.	Access is denied.
SV-02	Input Validation	SEC-02, NFR-09	Submit malformed input in registration/message fields.	Input is rejected or safely handled without unintended processing.
SV-03	Administrative Authorization	SEC-04	Attempt an administrative operation as an unauthorized user.	Operation is denied.
SV-04	Invalid Session	SEC-01, SEC-03	Attempt protected activity using an invalid or terminated session.	Protected activity is not permitted.

4. Requirement Coverage Summary

Requirement Group	Covered By	Coverage
Functional Requirements	TC-01 to TC-10	Registration, authentication, connections, messaging, users, validation, errors.
Performance / Response Time	TC-11	Measured message delivery and user-action response time.
Concurrency	TC-12	Multiple simultaneous clients.
Availability	TC-13	Service continuity and reconnect behavior.
Reliability	TC-14	Message delivery behavior.
Usability	TC-15	Basic user workflow.
Security / Input Validation	TC-16 to TC-18, SV-01 to SV-04	Validation, authentication, authorization.

5. Execution and Recording

During execution, the tester shall record the actual result, execution date, tester, environment, and Pass/Fail status for each case. Failed cases shall be associated with a defect or issue record where applicable.

6. Update to Existing Test Plan

The test cases in this document should be incorporated into Section 5 of the Test Plan. The security validation cases should remain in Section 5.1. The SRS traceability table should be updated to reference the final test-case IDs.

7. Acceptance Criteria

- At least 7–10 test cases are required; this update provides 18 primary cases plus 4 security validation cases.
- Both functional and non-functional requirements are covered.

- Security objectives and security requirements have explicit validation cases.
- Every test case has an SRS trace where applicable.
- Performance tests use measurable criteria rather than subjective assessment.